Business Problem:

A telecom company is losing customers every month, and acquiring a new customer costs far more than keeping an existing one. This project identifies who is churning and why, identifies customers at risk, and recommends retention actions to protect revenue.

Objectives:

1. To find the overall churn rate and the monthly revenue lost to churn.
2. To find which factors are linked to churn, such as contract type, tenure, internet service, payment method, and add-on services.
3. To locate the customer groups with the highest churn rate and revenue loss.
4. To suggest retention strategies.

Dataset Summary:

- **Source:** IBM Telco Customer Churn dataset (Kaggle)
- **Tables:** 1
- **Rows:** 7,043 customers, one row each
- **Columns:** 21 (23 after cleaning)
- **Target:** Churn (Yes/No). 1,869 customers (26.54%) churned.

Key Features

- **Identifier:** customerID
- **Demographics:** gender, SeniorCitizen, Partner, Dependents
- **Services:** PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
- **Account:** tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
- **Target:** Churn

Added During Cleaning

- ChurnFlag: churn as 1/0 for calculations
- TenureGroup: tenure buckets (0-12, 13-24, 25-48, 49-72 months)

Data Quality

- No duplicate rows
- 11 blank TotalCharges values (new customers with tenure 0), set to 0
- SeniorCitizen converted from 0/1 to No/Yes

Exploratory Data Analysis using Python:

It starts with data preparation and cleaning in Python:

Data Loading: Imported the dataset using Pandas.
Exploration: The dataset has only one table, so no joins were needed. The data was explored in Python to understand its structure and quality. df.head() was used to view the first rows, df.info() to

check the structure and data types, and df.describe(include="all") to get summary statistics for both numeric and text columns. df.isnull().sum() counted the missing values in each column, and df.duplicated().sum() counted the duplicate rows.

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...
count	7043	7043	7043.000000	7043	7043	7043.000000	7043	7043	7043	7043	...
unique	7043	2	NaN	2	2	NaN	2	3	3	3	...
top	7590-VHVEG	Male	NaN	No	No	NaN	Yes	No	Fiber optic	No	...
freq	1	3555	NaN	3641	4933	NaN	6361	3390	3096	3498	...
mean	NaN	NaN	0.162147	NaN	NaN	32.371149	NaN	NaN	NaN	NaN	...
std	NaN	NaN	0.368612	NaN	NaN	24.559481	NaN	NaN	NaN	NaN	...
min	NaN	NaN	0.000000	NaN	NaN	0.000000	NaN	NaN	NaN	NaN	...
25%	NaN	NaN	0.000000	NaN	NaN	9.000000	NaN	NaN	NaN	NaN	...
50%	NaN	NaN	0.000000	NaN	NaN	29.000000	NaN	NaN	NaN	NaN	...
75%	NaN	NaN	0.000000	NaN	NaN	55.000000	NaN	NaN	NaN	NaN	...
max	NaN	NaN	1.000000	NaN	NaN	72.000000	NaN	NaN	NaN	NaN	...

DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
7043	7043	7043	7043	7043	7043	7043	7043.000000	7043	7043
3	3	3	3	3	2	4	NaN	6531	2
No	No	No	No	Month-to-month	Yes	Electronic check	NaN	20.2	No
3095	3473	2810	2785	3875	4171	2365	NaN	11	5174
NaN	NaN	NaN	NaN	NaN	NaN	NaN	64.761692	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	30.090047	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	18.250000	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	35.500000	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	70.350000	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	89.850000	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	118.750000	NaN	NaN

Customers with missing TotalCharges (all have tenure 0):

	customerID	tenure
488	4472-LVYGI	0
753	3115-CZMZD	0
936	5709-LVOEQ	0
1082	4367-NUYAO	0
1340	1371-DWPAZ	0
3331	7644-OMVMY	0
3826	3213-VVOLG	0
4380	2520-SGTTA	0
5218	2923-ARZLG	0
6670	4075-WKNIU	0
6754	2775-SEFEE	0

Handling Missing values: Missing values in TotalCharges were imputed with 0, because they belonged to new customers with zero tenure who had not yet been billed.

Data Transformation and Feature Creation:

After cleaning, the data was transformed to make it consistent and easier to analyze in SQL and Power BI.

Standardization

- SeniorCitizen was recoded from 0/1 to No/Yes, to match the other yes/no columns in the dataset.

New Features

- ChurnFlag: created from Churn as 1 (churned) or 0 (stayed). This allows the churn rate to be calculated directly as the average of the column.
- TenureGroup: created by grouping tenure into four ranges: 0-12, 13-24, 25-48, and 49-72 months. This makes it easier to compare churn across customer lifecycle stages.

The final dataset has 23 columns and was saved as telco_clean.csv.

Database Integration: Connected Python script with postgresql and loaded the clean dataframe into the database for sql analysis.

Data Analysis Using SQL:

Structured analysis was performed in PostgreSQL to answer the key business questions:

Overall churn: This query calculates the total number of customers, the number of churned customers, and the overall churn rate (26.54%).

customers	churned_customer	overall_churn
Filter...	Filter...	Filter...
7043	1869	26.54

Revenue lost: Churned customers accounted for about $139K of monthly revenue, which is about 30.5% of total monthly revenue. Churn costs more revenue than its customer share (26.5%) because churned customers pay higher monthly charges on average.

revenu_lost
Filter...
139130.85

total_revenue	lost_revenue	lost_pct
Filter...	Filter...	Filter...
456116.60	139130.85	30.50

Churn By Contract type: Churn was analyzed by contract type. Month-to-month customers had the highest churn rate (about 43%), compared with about 11% for one-year contracts and about 3% for

two-year contracts. This shows that customers without a long-term commitment are far more likely to leave.

contract	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
Month-to-month	3875	1655	42.71
One year	1473	166	11.27
Two year	1695	48	2.83

Churn by tenure group: Churn was analyzed by tenure group. Customers in their first 12 months had the highest churn rate (about 48%), and churn fell steadily as tenure increased, to about 9% for customers with 49 to 72 months. This shows that customers are most likely to leave early, so retention efforts should focus on new customers.

tenuregroup	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
0-12	2186	1037	47.44
13-24	1024	294	28.71
25-48	1594	325	20.39
49-72	2239	213	9.51

Churn by internet service: Churn was analyzed by internet service. Fiber optic customers had the highest churn rate (about 42%), compared with about 19% for DSL and about 7% for customers with no internet service. Since fiber customers also pay the highest monthly charges, this suggests a possible problem with the value or quality of the fiber service.

internetservice	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
Fiber optic	3096	1297	41.89
DSL	2421	459	18.96
No	1526	113	7.40

Churn by payment method: Churn was analyzed by payment method. Customers paying by electronic check had the highest churn rate (about 45%), compared with about 15% to 19% for the other methods. Customers on automatic payments (bank transfer or credit card) churned the least, which suggests that moving customers to autopay could help retention.

paymentmethod	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
Electronic check	2365	1071	45.29
Mailed check	1612	308	19.11
Bank transfer (automatic)	1544	258	16.71
Credit card (automatic)	1522	232	15.24

Churn by Add-on Service: Churn was analyzed by add-on services. Customers without tech support, online security, or online backup churned at about 40% to 42%, compared with about 15% for customers who had tech support or online security and about 22% for those with online backup. Customers with no internet service churned the least (about 7%). This suggests that protection services are linked to higher retention, so bundling them could help reduce churn.

techsupport	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
No	3473	1446	41.64
Yes	2044	310	15.17
No internet service	1526	113	7.40

onlinesecurity	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
No	3498	1461	41.77
Yes	2019	295	14.61
No internet service	1526	113	7.40

onlinebackup	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
No	3088	1233	39.93
Yes	2429	523	21.53
No internet service	1526	113	7.40

Average Monthly Charges: Churned customers paid more per month on average (about $74) than retained customers (about $61). This suggests that higher-priced customers are more likely to leave, which fits the high churn among fiber optic customers.

churn	avg_monthly_ch...
Filter...	Filter...
No	61.27
Yes	74.44

Churn by Demographics: Churn was analyzed by demographic group. Senior citizens churned at about 42%, compared with about 24% for non-seniors. Customers with no partner churned at about

33%, compared with about 20% for customers with a partner. Customers with no dependents churned at about 31%, compared with about 15% for customers with dependents. Customers who live alone, have no dependents, or are senior citizens are therefore the higher-risk groups.

seniorcitizen	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
Yes	1142	476	41.68
No	5901	1393	23.61

partner	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
No	3641	1200	32.96
Yes	3402	669	19.66

dependents	customers	churned_customer	churn_pct
Filter...	Filter...	Filter...	Filter...
No	4933	1543	31.28
Yes	2110	326	15.45

High-risk segments: High-risk segments were identified by combining contract type, internet service, and payment method. The highest-risk segment was month-to-month customers with fiber optic service paying by electronic check. This segment had about 1,307 customers, a churn rate of about 60%, and about $68K of monthly revenue lost, which is about half of all revenue lost to churn. Targeting this group with retention offers would have the largest impact.

contract	internetservice	paymentmethod	customers	churned	churn_pct	monthly_revenu...
Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
Month-to-month	Fiber optic	Electronic check	1307	789	60.37	68281.50
Month-to-month	Fiber optic	Bank transfer (automatic)	327	149	45.57	13062.10
Month-to-month	Fiber optic	Credit card (automatic)	293	122	41.64	10731.15
Month-to-month	DSL	Electronic check	474	192	40.51	8776.45
Month-to-month	Fiber optic	Mailed check	201	102	50.75	8407.25
Month-to-month	DSL	Mailed check	367	113	30.79	5292.10
One year	Fiber optic	Electronic check	196	51	26.02	5149.65
Month-to-month	DSL	Credit card (automatic)	185	50	27.03	2474.80
One year	Fiber optic	Credit card (automatic)	150	23	15.33	2392.35
One year	Fiber optic	Bank transfer (automatic)	157	23	14.65	2344.35
Month-to-month	DSL	Bank transfer (automatic)	197	39	19.80	1823.90
Two year	Fiber optic	Bank transfer (automatic)	162	15	9.26	1580.15
Month-to-month	No	Mailed check	325	67	20.62	1345.05
One year	DSL	Credit card (automatic)	172	17	9.88	1133.00
Two year	Fiber optic	Electronic check	92	9	9.78	915.25
One year	DSL	Mailed check	137	14	10.22	894.65
One year	DSL	Bank transfer (automatic)	145	11	7.59	696.75
Two year	Fiber optic	Credit card (automatic)	154	6	3.90	655.20
One year	DSL	Electronic check	116	11	9.48	631.85
Two year	DSL	Credit card (automatic)	237	5	2.11	361.70
Month-to-month	No	Bank transfer (automatic)	65	13	20.00	265.95
Month-to-month	No	Electronic check	69	13	18.84	257.65
Two year	DSL	Electronic check	58	4	6.90	231.10
Two year	DSL	Bank transfer (automatic)	224	3	1.34	212.90
Month-to-month	No	Credit card (automatic)	65	6	9.23	129.20
One year	No	Bank transfer (automatic)	89	4	4.49	86.05
Two year	No	Credit card (automatic)	190	2	1.05	49.90
Two year	No	Mailed check	252	2	0.79	43.85
One year	No	Mailed check	164	2	1.22	39.60
Two year	No	Bank transfer (automatic)	178	1	0.56	19.75
One year	No	Credit card (automatic)	76	1	1.32	19.30
Two year	DSL	Mailed check	109	0	0.00	0

Churn by Number of Protection Add-ons: Churn was analyzed by the number of protection add-ons a customer has (online security, online backup, and tech support). Customers with no add-ons had the highest churn rate (about 54%), and churn fell steadily as the number of add-ons increased, to about 7% for customers with all three. This suggests that protection services are linked to higher retention, so bundling or offering them free to at-risk customers could reduce churn.

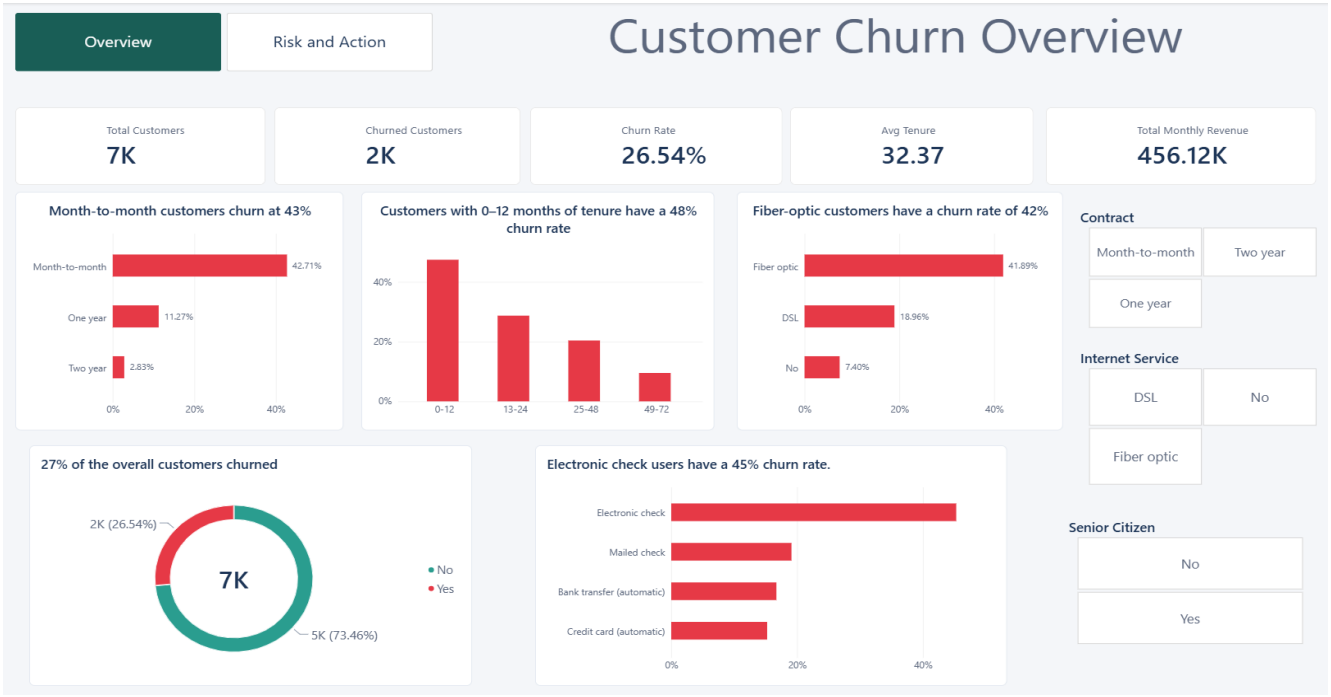
addon_count	customers	churned	churn_pct
Filter...	Filter...	Filter...	Filter...
0	1713	925	54.00
1	1805	581	32.19
2	1310	203	15.50
3	689	47	6.82

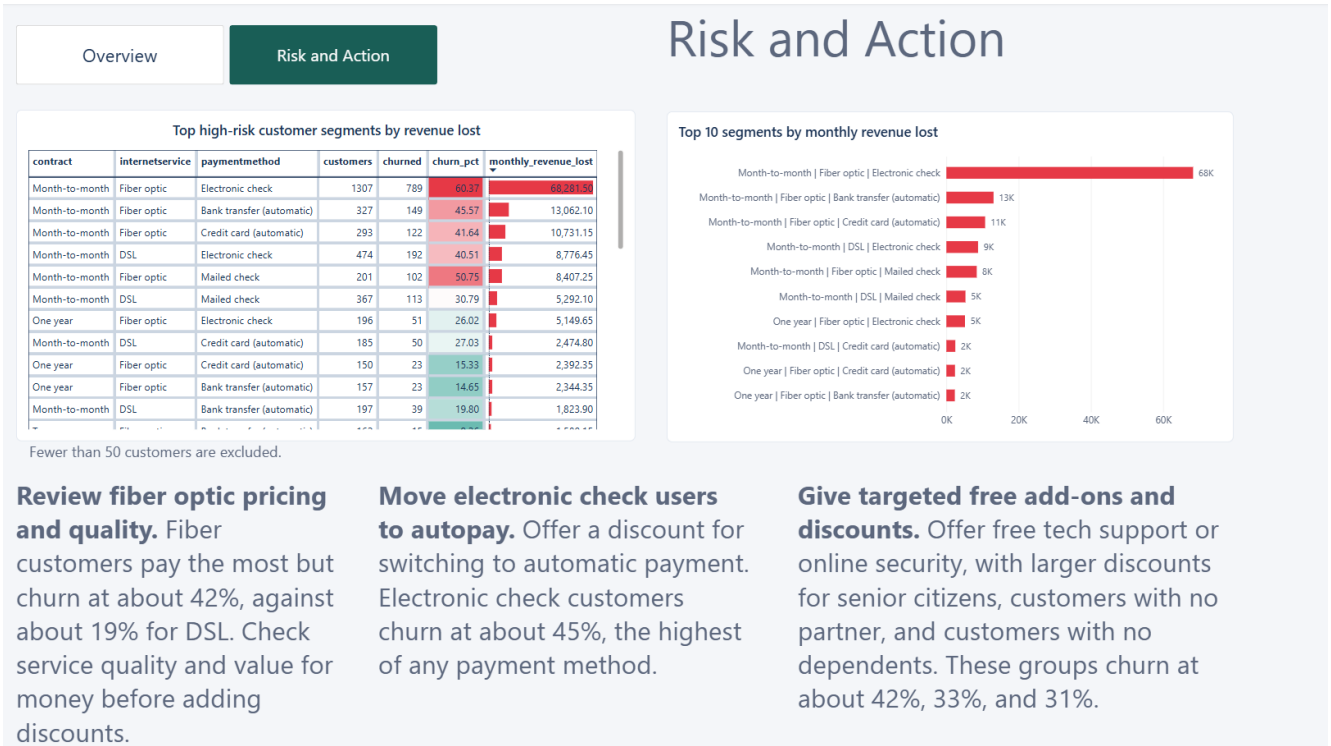
Customer Value: Churned and retained customers were compared on tenure and total charges. Churned customers stayed about 18 months on average, compared with about 37 months for retained customers. Their average total charges were also much lower (about $1,531 vs about $2,550). This shows that customers leave early, before they generate much lifetime revenue, so retention efforts should focus on the first year.

avg_retained_te...	avg_churned_te...	avg_retained_to...	avg_churned_tot...
Filter...	Filter...	Filter...	Filter...
37.57	17.98	2549.91	1531.80

Dashboards in Power BI:

Finally, an interactive two-page dashboard was built in Power BI to present the insights visually. It connects directly to the PostgreSQL database.





Fewer than 50 customers are excluded.

Review fiber optic pricing and quality. Fiber customers pay the most but churn at about 42%, against about 19% for DSL. Check service quality and value for money before adding discounts.

Move electronic check users to autopay. Offer a discount for switching to automatic payment. Electronic check customers churn at about 45%, the highest of any payment method.

Give targeted free add-ons and discounts. Offer free tech support or online security, with larger discounts for senior citizens, customers with no partner, and customers with no dependents. These groups churn at about 42%, 33%, and 31%.

Business Recommendations:

1. Move month-to-month customers to longer contracts

- **Finding:** Month-to-month customers churn at about 43%, against about 3% for two-year contracts. The top risk segment (month-to-month, fiber, electronic check) churns at about 60% and loses about $68K a month.
- **Action:** Offer contract-upgrade discounts, starting with that segment.
- **Expected impact:** The largest revenue saving of the three.

2. Encourage autopay and review fiber service

- **Finding:** Electronic check customers churn at about 45%, against about 15% to 19% for other payment methods. Fiber customers churn at about 42%, against about 19% for DSL, despite paying the highest charges.
- **Action:** Offer a discount for switching to automatic payment. Review fiber pricing and service quality to see why these customers leave.
- **Expected impact:** Lower churn among the payment and service groups with the highest rates.

3. Protect new customers with free add-ons

- **Finding:** Customers in their first 12 months churn at about 48%, and churn falls as the number of add-ons rises.
- **Action:** Give new customers free tech support or online security for the first few months.
- **Expected impact:** Better early retention, since customers leave before generating much lifetime revenue.